

# **Lecture 27**

**STAT 109: Introductory Biostatistics**

# Lecture 27

## Two categorical variables: contingency tables, proportions, and stacked bar charts

### Learning objectives

- Build a two-way table (contingency table) of counts for two categorical variables.
- Convert counts to proportions three ways: joint (of the grand total), within rows, and within columns—and say which question each answers.
- Make stacked bar charts in ggplot2 (`geom_col()` with `fill`); interpret stacked counts vs 100% stacked (`position = "fill"`).
- Reproduce tables and plots using the tidyverse (`count()`, `group_by()`, `mutate()`, `pivot_wider()`).

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### Why two categorical variables together

Each observation has a level of variable A (e.g. species) and a level of variable B (e.g. habitat). We summarize how the combinations co-occur: counts in each cell of a table, then graphs can show these counts or proportions at a glance.

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### Small example by hand (two binary variables)

**Setting.** Birds counted in two habitats: **Urban** or **Rural**. Species recorded as **Robin** or **Sparrow** (ignore other species here).

**Counts:**

|              | Urban | Rural | Row total                   |
|--------------|-------|-------|-----------------------------|
| Robin        | 8     | 2     | 10                          |
| Sparrow      | 4     | 6     | 10                          |
| Column total | 12    | 8     | <b>20</b> = grand total $N$ |

So there are **20** birds. The **joint** count in each cell is a count of birds with that (**species, habitat**) pair.

### Joint proportions (of the grand total)

Each cell count divided by  $N = 20$  answers: *What fraction of **all** birds fall in this cell?*

- Robin & Urban:  $8/20 = 0.40$
- Robin & Rural:  $2/20 = 0.10$
- Sparrow & Urban:  $4/20 = 0.20$
- Sparrow & Rural:  $6/20 = 0.30$

Check:  $0.40 + 0.10 + 0.20 + 0.30 = 1$ .

## Row proportions (conditional on species)

Within each **species**, divide by that row's total (10). Answers: *Among Robins only, what fraction were Urban vs Rural?* (and similarly for Sparrows.)

**Robin:** Urban  $8/10 = 0.80$ , Rural  $2/10 = 0.20$ .

**Sparrow:** Urban  $4/10 = 0.40$ , Rural  $6/10 = 0.60$ .

Each row sums to 1.

## Column proportions (conditional on habitat)

Within each **habitat**, divide by that column's total. Answers: *Among Urban birds, what fraction were Robin vs Sparrow?*

**Urban:** Robin  $8/12 \approx 0.67$ , Sparrow  $4/12 \approx 0.33$  (column total 12).

**Rural:** Robin  $2/8 = 0.25$ , Sparrow  $6/8 = 0.75$  (column total 8).

Each column sums to 1.

Note: The same four counts support three different stories depending on whether you normalize by everything (joint or "grand"), by rows (species), or by columns (habitat). Be explicit about which denominator you use when you write a sentence or label a plot.

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## Example 1: The bird table in R's tidyverse

We store one row per bird: variables `species` and `habitat`.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

birds <- tibble(
  species = c(rep("Robin", 10), rep("Sparrow", 10)),
  habitat = c(rep(c("Urban", "Rural"), c(8, 2)), rep(c("Urban", "Rural"), c(4, 6)))
) |>
  mutate(
    species = factor(species, levels = c("Robin", "Sparrow")),
    habitat = factor(habitat, levels = c("Urban", "Rural"))
  )
```

## Two-way table of counts

```
birds |>
  count(species, habitat) |>
  pivot_wider(names_from = habitat, values_from = n, values_fill = 0)
```

```
# A tibble: 2 x 3
  species Urban Rural
  <fct>   <int> <int>
1 Robin         8     2
2 Sparrow        4     6
```

## Joint, row, and column proportions (long format)

```
joint <- birds |>
  count(species, habitat) |>
  mutate(prop_joint = n / sum(n))

row_prop <- birds |>
  count(species, habitat) |>
  group_by(species) |>
  mutate(prop_row = n / sum(n)) |>
  ungroup()

col_prop <- birds |>
  count(species, habitat) |>
  group_by(habitat) |>
  mutate(prop_col = n / sum(n)) |>
  ungroup()

joint
```

```
# A tibble: 4 x 4
  species habitat     n prop_joint
  <fct>   <fct>   <int>     <dbl>
1 Robin   Urban      8       0.4
2 Robin   Rural      2       0.1
3 Sparrow Urban      4       0.2
4 Sparrow Rural      6       0.3
```

```
row_prop
```

```
# A tibble: 4 x 4
  species habitat     n prop_row
  <fct>   <fct>   <int>     <dbl>
1 Robin   Urban      8       0.8
2 Robin   Rural      2       0.2
3 Sparrow Urban      4       0.4
4 Sparrow Rural      6       0.6
```

```
col_prop
```

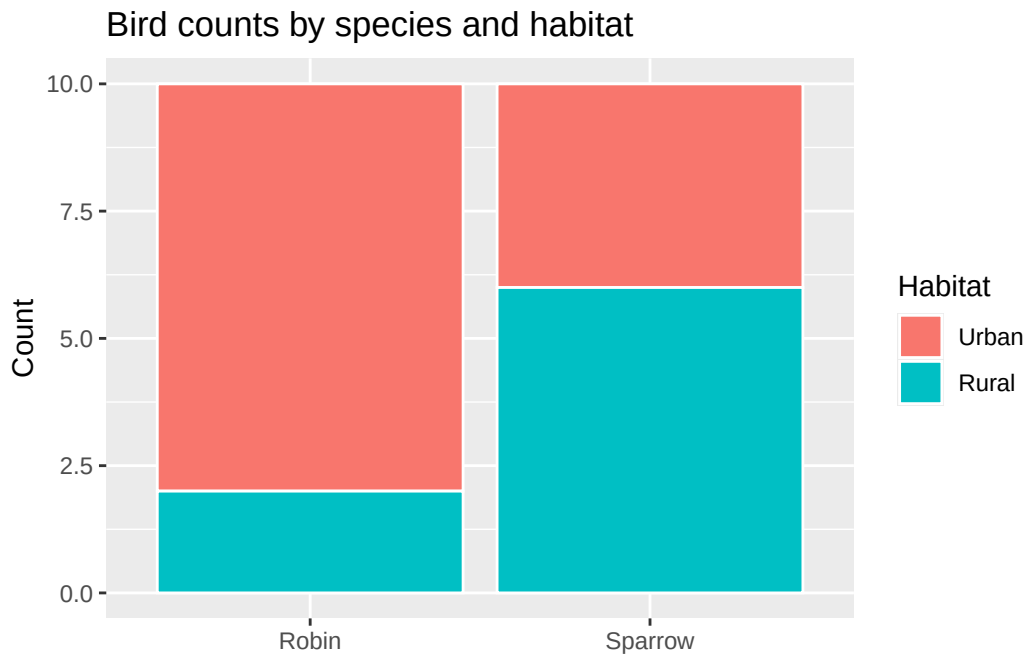
```
# A tibble: 4 x 4
  species habitat     n prop_col
  <fct>   <fct>   <int>     <dbl>
1 Robin   Urban      8    0.667
2 Robin   Rural      2    0.25
3 Sparrow Urban      4    0.333
4 Sparrow Rural      6    0.75
```

## Stacked bar chart of counts

Bars show **species**; segments show **habitat**; height of each segment is the **cell count**.

```
birds |>
  count(species, habitat) |>
```

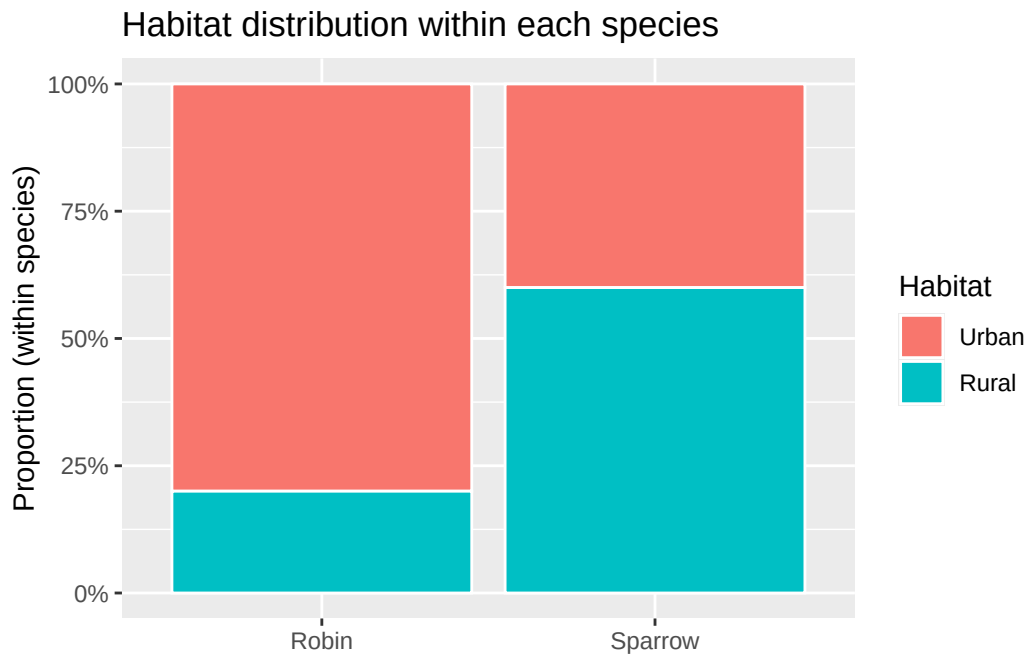
```
ggplot(aes(x = species, y = n, fill = habitat)) +
  geom_col(color = "white") +
  labs(
    title = "Bird counts by species and habitat",
    x = NULL,
    y = "Count",
    fill = "Habitat"
  )
```



### Stacked bar chart with 100% within each species (row proportions)

`position = "fill"` stretches each bar to height 1; segment **heights** match **row proportions** (`prop_row`). Good for comparing **habitat mix** between species when raw counts differ.

```
birds |>
  count(species, habitat) |>
  ggplot(aes(x = species, y = n, fill = habitat)) +
  geom_col(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::label_percent()) +
  labs(
    title = "Habitat distribution within each species",
    x = NULL,
    y = "Proportion (within species)",
    fill = "Habitat"
  )
```



Notice that you swap what is on the **x-axis** and what is **fill** if the research question reads more naturally the other way (e.g. habitat on the x-axis, species as fill).

## Example 2: mpg: class vs drv (several categories each)

Each row is a car model. We cross **vehicle class** and **drive type**.

```
data("mpg", package = "ggplot2")

mpg |>
  count(class, drv) |>
  pivot_wider(names_from = drv, values_from = n, values_fill = 0)
```

```
# A tibble: 7 x 4
  class      r `4`   f
  <chr>    <int> <int> <int>
1 2seater     5     0     0
2 compact     0    12    35
3 midsize     0     3    38
4 minivan     0     0    11
5 pickup      0    33     0
6 subcompact  9     4    22
7 suv        11    51     0
```

Row proportions: among cars in a given **class**, what fraction have each **drv**?

```
mpg |>
  count(class, drv) |>
  group_by(class) |>
  mutate(prop_within_class = n / sum(n)) |>
  ungroup() |>
  print(n = 20)
```

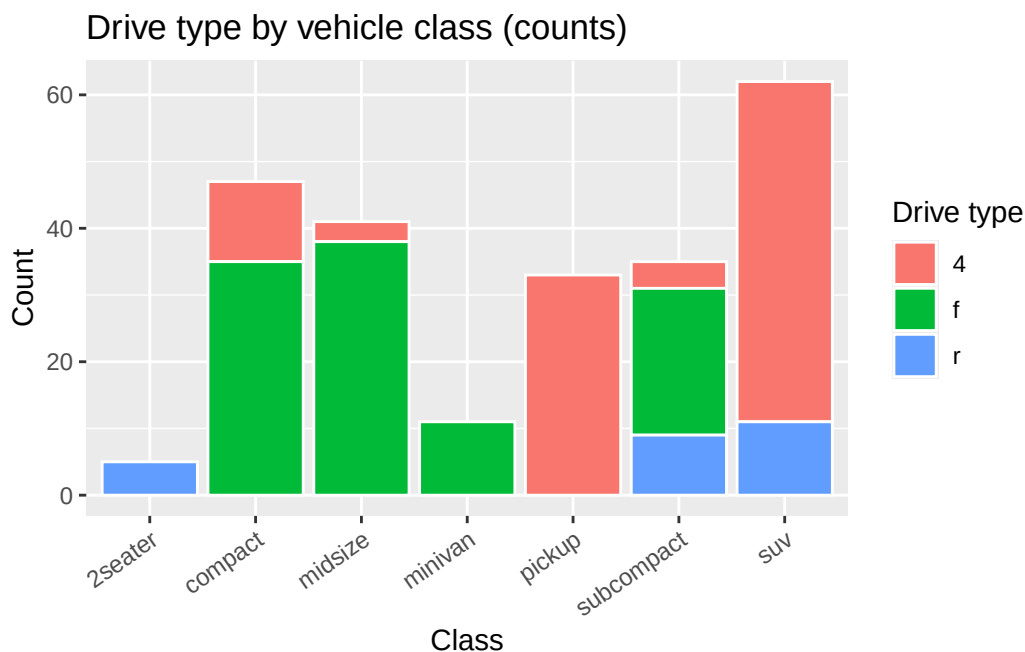
# A tibble: 12 x 4

|    | class      | drv   | n     | prop_within_class |
|----|------------|-------|-------|-------------------|
|    | <chr>      | <chr> | <int> | <dbl>             |
| 1  | 2seater    | r     | 5     | 1                 |
| 2  | compact    | 4     | 12    | 0.255             |
| 3  | compact    | f     | 35    | 0.745             |
| 4  | midsize    | 4     | 3     | 0.0732            |
| 5  | midsize    | f     | 38    | 0.927             |
| 6  | minivan    | f     | 11    | 1                 |
| 7  | pickup     | 4     | 33    | 1                 |
| 8  | subcompact | 4     | 4     | 0.114             |
| 9  | subcompact | f     | 22    | 0.629             |
| 10 | subcompact | r     | 9     | 0.257             |
| 11 | suv        | 4     | 51    | 0.823             |
| 12 | suv        | r     | 11    | 0.177             |

Stacked bars (counts and 100% stacked):

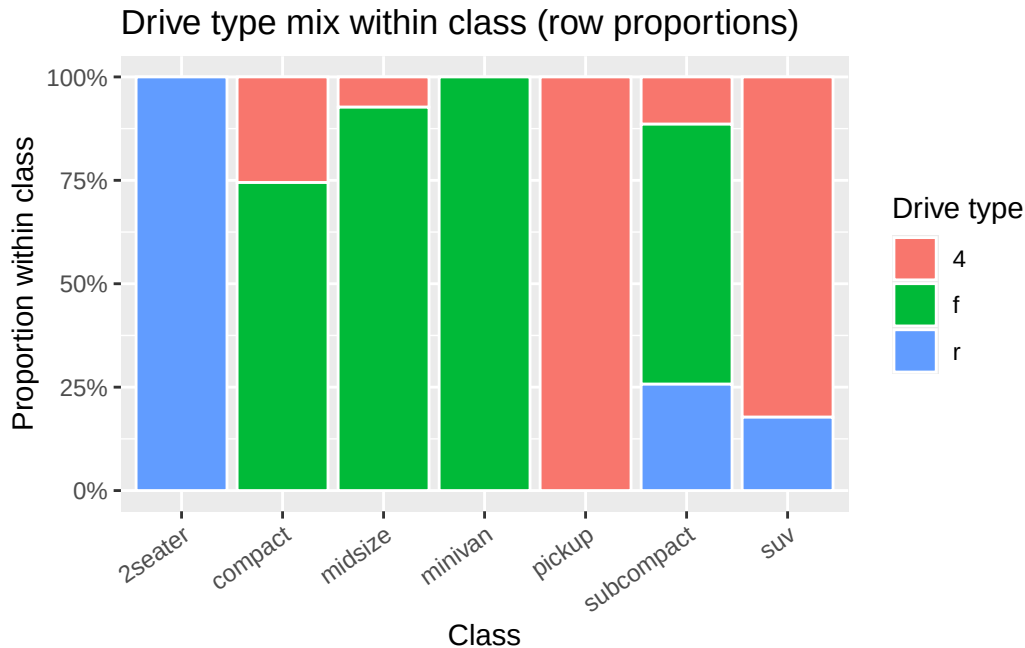
```
mpg_ct <- mpg |> count(class, drv)

ggplot(mpg_ct, aes(x = class, y = n, fill = drv)) +
  geom_col(color = "white") +
  labs(
    title = "Drive type by vehicle class (counts)",
    x = "Class",
    y = "Count",
    fill = "Drive type"
  ) +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))
```



```
ggplot(mpg_ct, aes(x = class, y = n, fill = drv)) +
  geom_col(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::label_percent()) +
```

```
labs(
  title = "Drive type mix within class (row proportions)",
  x = "Class",
  y = "Proportion within class",
  fill = "Drive type"
) +
theme(axis.text.x = element_text(angle = 35, hjust = 1))
```



### Example 3: TidyTuesday Palmer Penguins: species vs island

I load the published TidyTuesday file from GitHub with `read_csv()`. One could also download the CSV once and use a local path instead.

```
penguins <- read_csv(
  "https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/2020-07-28/penguins.csv",
  show_col_types = FALSE
)

penguins |>
  count(species, island) |>
  pivot_wider(names_from = island, values_from = n, values_fill = 0)
```

```
# A tibble: 3 x 4
  species   Biscoe Dream Torgersen
  <chr>     <int> <int>     <int>
1 Adelie      44    56        52
2 Chinstrap    0    68         0
3 Gentoo    124     0         0
```

**Column proportions:** among birds on a given **island**, what fraction belong to each **species**?



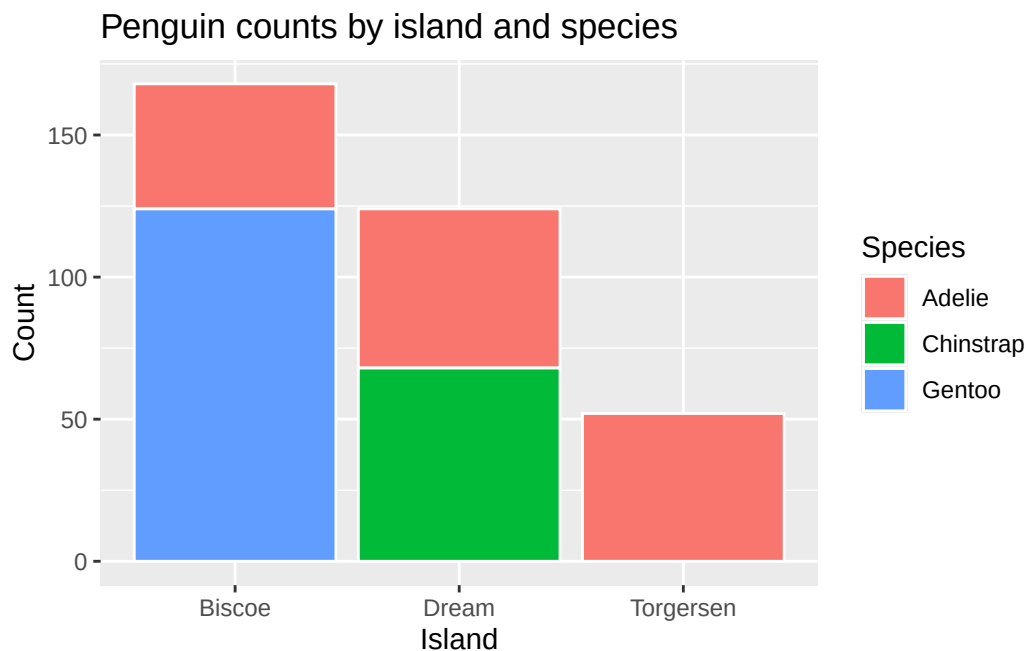
```
penguins |>
  drop_na(species, island) |>
  count(species, island) |>
  group_by(island) |>
  mutate(prop_within_island = n / sum(n)) |>
  ungroup()
```

```
# A tibble: 5 x 4
  species   island      n prop_within_island
  <chr>     <chr>   <int>         <dbl>
1 Adelie   Biscoe     44         0.262
2 Adelie   Dream     56         0.452
3 Adelie   Torgersen  52          1
4 Chinstrap Dream    68         0.548
5 Gentoo   Biscoe    124         0.738
```

Stacked bars: counts and **column-style** emphasis: put **island** on the x-axis and **species** as **fill** so each bar is an island total and segments show species mix on that island.

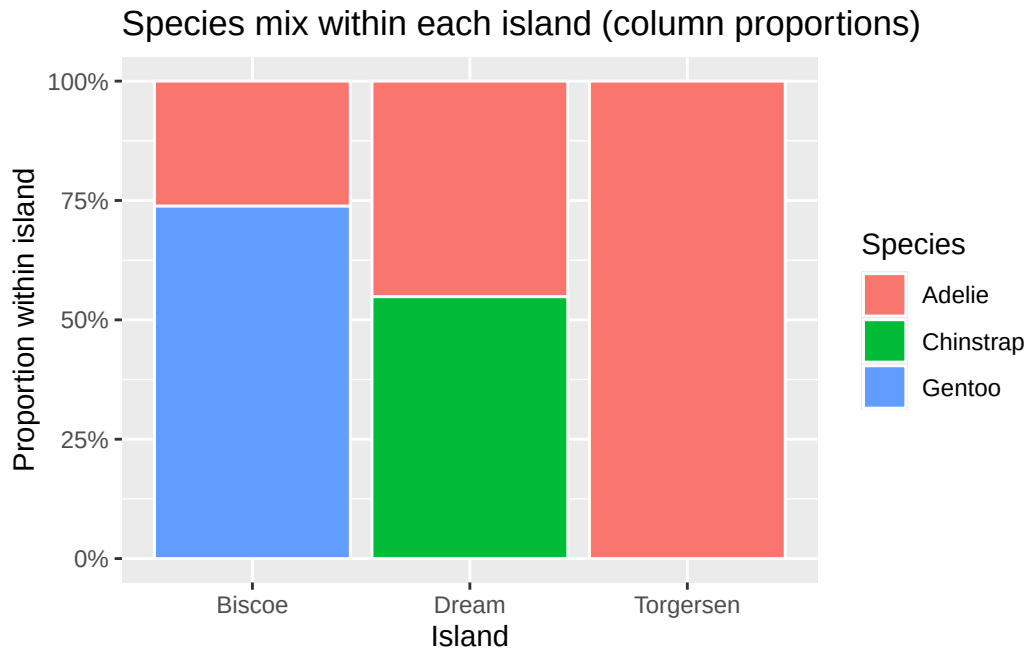
```
pg <- penguins |> drop_na(species, island) |> count(species, island)

ggplot(pg, aes(x = island, y = n, fill = species)) +
  geom_col(color = "white") +
  labs(
    title = "Penguin counts by island and species",
    x = "Island",
    y = "Count",
    fill = "Species"
  )
```



```
ggplot(pg, aes(x = island, y = n, fill = species)) +
  geom_col(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::label_percent()) +
```

```
labs(
  title = "Species mix within each island (column proportions)",
  x = "Island",
  y = "Proportion within island",
  fill = "Species"
)
```



## Summary

For two categorical variables, start with counts in a two-way table, then compute proportions with a clear denominator: joint ( $N$  everywhere), row (within levels of the variable on the rows), or column (within levels of the variable on the columns). Stacked bar charts display the same structure: stacked counts show how many; position = “fill” shows fractions within each bar (match the fraction definition to rows or columns by choosing  $x$  and fill thoughtfully).